Expansion in finite simple groups of Lie type

Terence Tao · 10 September 2013

https://terrytao.wordpress.com/2013/09/10/expansion-in-finite-simple-groups-of-lie-type/

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Emmanuel Breuillard, Ben Green, Bob Guralnick, and I have just uploaded to the arXiv our joint paper "Expansion in finite simple groups of Lie type". This long-delayed paper (announced way back in 2010!) is a followup to our previous paper in which we showed that, with one possible exception, generic pairs of elements of a simple algebraic group (over an uncountable field) generated a free group which was strongly dense in the sense that any nonabelian subgroup of this group was Zariski dense. The main result of this paper is to establish the analogous result for finite simple groups of Lie type (as defined in the previous blog post) and bounded rank, namely that almost all pairs a, b of elements of such a group generate a Cayley graph which is a (two-sided) expander, with expansion constant bounded below by a quantity depending on the rank of the group. (Informally, this means that the random walk generated by a, b spreads out in logarithmic time to be essentially uniformly distributed across the group, as opposed for instance to being largely trapped in an algebraic subgroup. Thus if generic elements did not generate a strongly dense group, one would probably expect expansion to fail.)

There are also some related results established in the paper. Firstly, as we discovered after writing our first paper, there was one class of algebraic groups for which our demonstration of strongly dense subgroups broke down, namely the Sp_4 groups in characteristic three. In the current paper we provide in a pair of appendices a new argument that covers this case (or more generally, Sp_4 in odd characteristic), by first reducing to the case of affine groups $k^2 \rtimes SL_2(k)$ (which can be found inside Sp_4 as a subgroup) and then using a ping-pong argument (in a p-adic metric) in the latter context.

Secondly, we show that the distinction between one-sided expansion and two-sided expansion (see this set of lecture notes of mine for definitions) is erased in the context of Cayley graphs of bounded degree, in the sense that such graphs are one-sided expanders if and only if they are two-sided expanders (perhaps with slightly different expansion constants). The argument turns out to be an elementary combinatorial one, based on the "pivot" argument discussed in these lecture notes of mine.

Now to the main result of the paper, namely the expansion of random Cayley graphs. This result had previously been established for SL_2 by Bourgain and Gamburd, and Ben, Emmanuel and I had used the Bourgain-Gamburd method to achieve the same result for Suzuki groups. For the other finite simple groups of Lie type, expander graphs had been constructed by Kassabov, Lubotzky, and Nikolov, but they required more than two generators, which were placed deterministically rather than randomly. (Here, I am skipping over a large number of other results on expanding Cayley graphs; see this survey of Lubotzsky for a fairly recent summary of developments.) The current paper also uses the "Bourgain-Gamburd machine", as discussed in these lecture notes of mine, to demonstrate expansion. This machine shows how expansion of a Cayley graph follows from three basic ingredients, which we state informally as follows:

- Non-concentration (A random walk in this graph does not concentrate in a proper subgroup);
- Product theorem (A medium-sized subset of this group which is not trapped in a proper subgroup will expand under multiplication); and
- Quasirandomness (The group has no small non-trivial linear representations).

Quasirandomness of arbitrary finite simple groups of Lie type was established many years ago (predating, in fact, the introduction of the term "quasirandomness" by Gowers for this property) by Landazuri-Seitz and Seitz-Zalesskii, and the product theorem was already established by Pyber-Szabo and independently by Breuillard, Green, and myself. So the main problem is to establish non-concentration: that for a

random Cayley graph on a finite simple group G of Lie type, random walks did not concentrate in proper subgroups.

The first step was to classify the proper subgroups of G. Fortunately, these are all known; in particular, such groups are either contained in proper *algebraic* subgroups of the algebraic group containing G (or a bounded cover thereof) with bounded complexity, or are else arising (up to conjugacy) from a version $G(F')$ of the same group $G = G(F)$ associated to a proper subfield F' of the field F respectively; this follows for instance from the work of Larsen and Pink, but also can be deduced using the classification of finite simple groups, together with some work of Aschbacher, Liebeck-Seitz, and Nori. We refer to the two types of subgroups here as "structural subgroups" and "subfield subgroups".

To preclude concentration in a structural subgroup, we use our previous result that generic elements of an algebraic group generate a strongly dense subgroup, and so do not concentrate in any algebraic subgroup. To translate this result from the algebraic group setting to the finite group setting, we need a Schwarz-Zippel lemma for finite simple groups of Lie type. This is straightforward for Chevalley groups, but turns out to be a bit trickier for the Steinberg and Suzuki-Ree groups, and we have to go back to the Chevalley-type parameterisation of such groups in terms of (twisted) one-parameter subgroups, that can be found for instance in the text of Carter; this "twisted Schwartz-Zippel lemma" may possibly have further application to analysis on twisted simple groups of Lie type. Unfortunately, the Schwartz-Zippel estimate becomes weaker in twisted settings, and particularly in the case of triality groups ${}^3D_4(q)$, which require a somewhat *ad hoc* additional treatment that relies on passing to a simpler subgroup present in a triality group, namely a central product of two different SL_2's.

To rule out concentration in a conjugate of a subfield group, we repeat an argument we introduced in our Suzuki paper and pass to a matrix model and analyse the coefficients of the characteristic polynomial of words in this Cayley graph, to prevent them from concentrating in a subfield. (Note that these coefficients are conjugation-invariant.)

Comments (10)

HUMBERTO · 10 September 2013

INTERESANTE LA CONEXIÓN

Anonymous · 10 September 2013

Typographical comment: For maps, one should use "\colon" in stead of ":" to get the correcty spacing.

> **Anonymous** · 10 September 2013
>
> (I'm talking about the paper on arXiv.)
> *[Thanks, this will be fixed in the next version of the ms. -T.]*

Anonymous · 11 September 2013

Comments to the bibliography:
* [12]: The "SL" should not be in italic mode. (Missing use of a LaTeX macro maybe?)
* [53]: The "GL" should not be in italic mode. (Missing use of a LaTeX macro maybe?)
* [12]–[17], [21], [25], [30], [32], [38], [43], [45], [47], [49], [57], [65]–[66]: When typing a page range, an en-dash (–) should be used instead of a hyphen (-).
* [41]: A space is needed after "pp.".
* [33], [39], [54], [56], [61], [67]: A full stop is missing at the very end of the references.
[Thanks, these will be fixed in the next revision of the ms. -T.]

Anonymous · 30 October 2013

Suggestions to Section 1 (but I'm not sure they are correct):
* P. 3, two lines below Theorem 1.2: "Lie type. See Theorem 8.3 there." –> "Lie type; see Theorem 8.3."
* P. 3, Comment (i), l. -1: "Petersen" –> "Petersson"
* P. 5, second remark, l. 3: "Lie type, and $a, b \in G$" –> "Lie type, $a, b \in G$"
* P. 5, second remark, l. -6: "SL_2" –> "SL_2"
[Thanks, these will be fixed in the next revision of the ms. -T.]

Anonymous · 30 October 2013

Suggestions to Section 2:
* P. 8, three lines after Proporsition 2.1: "Proposition" –> "proposition"
* P. 8, l. -1: Should "of Γ" be ommited? Isn't it superfluous?
* P. 9, the remark, l. 9: "Zariski dense" –> "Zariski-dense"
* P. 9, l. -3: "Zariski dense" –> "Zariski-dense"

> **Anonymous** · 30 October 2013
>
> "ommited" –> "omitted"

Anonymous · 30 October 2013

Suggestions to Section 3:
* P. 11, two lines before Proporsition 3.1: "$A \cdot A := \{a, b : a, b \in A\}$" –> "$A \cdot A := \{a \cdot b : a, b \in A\}$"
(P.S. Remember to use "\coloneqq" from the mathtools package for a proper `:='.)
* P. 12, four lines after Proporsition 3.1: "nonconcentration" –> "non-concentration"

Sean Eberhard · 28 January 2019

Dear Terry,

Any idea what the rank dependence is like? For example suppose I'm interested in $\mathrm{SL}(n,p)$ with $n \approx \log p$. Can I hope that these methods (in particular Larsen–Pink) might be effective?
Best,
Sean



Terence Tao · 28 January 2019

I think Emmanuel Breuillard has some unpublished work in this direction (possibly joint with my former student, Yilong Yang), but you would have to contact them directly about this.